

# Depth-Enabled Inspection of Medical Videos

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## Abstract

Cataract surgery is the most frequently performed surgical procedure worldwide, involving the replacement of a patient's clouded eye lens with a synthetic intraocular lens to restore visual acuity. Although typically brief, the operation consists of distinct phases that demand precision and extensive training, traditionally constrained by the limitations of real-time observation under a microscope. To enhance learning and procedural accuracy, modern advancements in stereoscopic video capture and head-mounted displays (HMDs) offer a promising solution. This paper demonstrates the application of stereoscopic cataract surgery videos, visualized through Apple Vision Pro (AVP) and Meta Quest 3, to provide immersive 3D perspectives that enhance depth perception and spatial awareness. An expert evaluation study with experienced surgeons indicates that stereoscopic visualization significantly improves comprehension of spatial relationships and procedural maneuvers, suggesting its potential to revolutionize surgical education and real-time guidance in ophthalmic surgery.

## Keywords

virtual reality, visualization techniques, medical videos

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## 1 Introduction

Cataract surgery [5] is the most frequently performed surgical procedure worldwide. Conducted under a microscope, this intervention involves replacing a patient's clouded eye lens with a synthetic intraocular lens, typically restoring full visual acuity. Although the operation usually lasts only a few minutes, it comprises ten distinct phases: *incision*, *viscous agent injection*, *rhexis*, *hydrodissection*, *phacoemulsification* (the longest and most critical step), *irrigation/aspiration*, *capsule polishing*, *lens implantation*, *viscous agent removal*, and *tonifying and antibiotics* [6].

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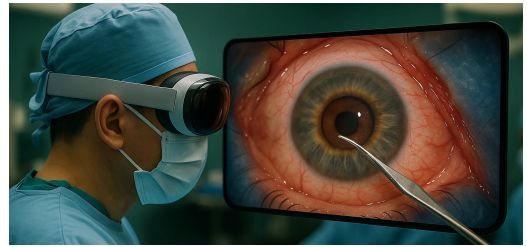


Figure 1: Stereoscopic visualization of cataract surgery.

Despite its routine nature, cataract surgery presents considerable challenges and demands extensive training and manual dexterity. Surgeons must master the precise manipulation with tiny instruments, and this requires intense training and the observation of many surgeries performed by expert surgeons. Unfortunately, training opportunities are limited by the conventional microscope setup, which allows only one observer to follow the procedure in real-time. To address this limitation, many hospitals now routinely record surgical procedures and use the videos for educational purposes [1]. However, a common 2D video camera, as currently widely used in ophthalmology, does not capture detailed depth information, which is crucial to an ophthalmic surgeon, though, and is clearly visible via the microscope. Therefore, newer equipment offers the opportunity of stereoscopic video recordings [2] with 3D displays in the operating room (OR). Unfortunately, this also limits the usage scenario for training, since only a small group of trainees can watch it, and with auto-stereoscopic displays, the quality heavily depends on the observation angle.

With advancements in head-mounted displays (HMDs) and support for stereoscopic videos, a new frontier opens for immersive training. In ophthalmology, HMDs offer medical students and professionals a near first-person view of complex operations, narrowing the gap between theory and practice. They can be used at any position/angle and even outside of the OR environment, and allow for much more frequent and precise surgery observations and inspections. This evolution in surgical video technology can enhance surgical training and lead to higher procedural safety [10, 11].

In this demo, as shown in Fig. 1, we present stereoscopic video inspection for cataract surgeries with Apple Vision Pro (AVP) and Meta Quest 3, demonstrating the enhanced depth perception and immersive experience enabled by modern HMDs. To assess this approach, we conducted an expert review with ophthalmic surgeons through a subjective test. The results show that experts recognize the benefits of stereoscopic visualization for surgical procedures and training. Surgeons noted improved spatial awareness and depth

perception, both crucial for cataract operations. Additionally, the immersive experience was highlighted as valuable for education, offering a near-real-world perspective beyond traditional 2D videos. These findings suggest that stereoscopic video technology could enhance learning and procedural accuracy in surgical training.

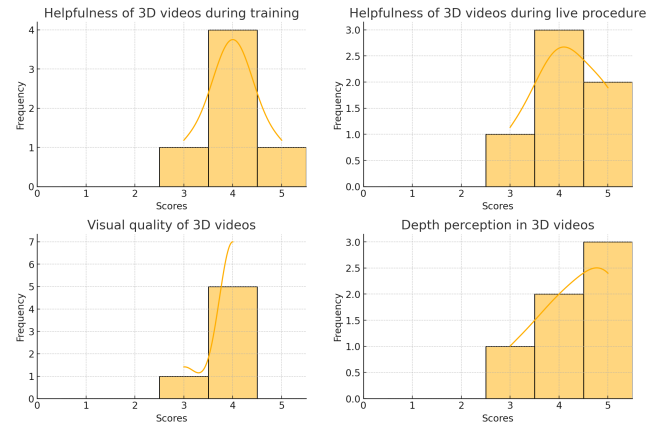
## 2 Stereoscopic Surgery Videos

Stereoscopic video technology captures visual information from two slightly different perspectives, simulating human depth perception [3, 9]. When viewed through an advanced 3D display, it enables 3D scene reconstruction, enhancing spatial awareness critical for medical training and surgical procedures [4, 7]. The latest HMDs, equipped with high-resolution displays and advanced rendering capabilities, further elevate the experience, ensuring smooth, depth-accurate visualization. This convergence of optimized capture, efficient compression, and high-performance displays has transformed stereoscopic video into a practical tool for immersive medical education, telemedicine, and interactive simulations, greatly improving spatial perception and training effectiveness.

In this demo, we use stereoscopic cataract surgery videos captured with a Leica Proveo 8 microscope, equipped with HD cameras (30 fps), with a focal distance (between lens and eye) of 175mm, focal length  $f=200$ mm, and a 3 degree angle of both cameras that use a separation distance of 24mm to achieve the necessary stereoscopic effect. The raw video was encoded using x265 v4.1 with support for MV-HEVC [8]. This setup allows for enhanced depth perception and realistic visualization during playback, providing an immersive experience that closely resembles the real-world surgical environment. We conducted an expert review with experienced surgeons to evaluate the effectiveness of stereoscopic imaging in cataract surgery [12–14].

## 3 Experimental Study

To evaluate the practicality and user experience of stereoscopic cataract surgery videos, we conducted a user study involving surgical professionals. The videos were presented on AVP, which were mirrored to a laptop screen. This mirroring not only facilitated easier training for surgeons unfamiliar with HMD technology but also enabled instructors to guide them through the setup process. For participants who wore prescription glasses and experienced difficulties with the fit of the AVP, we provided Meta Quest 3 as an alternative. Similar to the AVP setup, the Meta Quest 3 was also mirrored on a laptop, allowing real-time calibration of interpupillary distance (IPD) and easing the learning curve for using the device. This dual-setup approach ensured accessibility for all participants and enabled trainers to monitor and support the immersive viewing experience effectively. Participants were given the freedom to watch the stereoscopic surgery videos for as long as they desired and to explore multiple recordings at their own pace. This unrestricted viewing time was designed to allow surgeons to immerse themselves fully in the 3D content, appreciating the enhanced depth perception and spatial awareness provided by the HMDs. Following the viewing session, participants were asked to complete a questionnaire to assess their experience and perceived



**Figure 2: Mean Opinion Scores (MOS) for the evaluation of stereoscopic 3D videos in cataract surgery.**

value of the stereoscopic videos (during training and a live procedure). We also asked about the perceived visual quality and depth in the videos.

Six cataract surgery experts participated in the study. The ratings were collected using a 5-point Likert scale, ranging from 1 (very poor) to 5 (excellent). To evaluate the overall experience, we computed the Mean Opinion Scores (MOS) for each question. The resulting MOS values are visualized in Fig. 2. The results demonstrate the clear benefits of using an HMD with stereoscopic videos in enhancing depth perception and visual clarity for post-operative cataract surgery training, but also for usage during a live procedure (e.g., to better follow the depth-based actions of the observed surgeon). In contrast to common 2D videos, the surgeons agree that the depth perception was significantly enhanced, although the video quality itself could be improved (e.g., higher resolution and better alignment between left and right views).

## 4 Conclusions

In this paper, we presented the application of stereoscopic videos for cataract surgery training and evaluation using modern HMDs. Our study, involving six experienced cataract surgeons, demonstrated the effectiveness of 3D visualization in enhancing depth perception and spatial awareness – critical factors for surgical precision. The collected Mean Opinion Scores (MOS) reflected strong satisfaction with the visual quality and perceived helpfulness. These findings suggest that stereoscopic video technology, coupled with advanced HMDs, holds significant potential for improving surgical training experiences and bridging the gap between simulation and real-world practice.

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## References

- [1] Negin Ghamsarian, Hadi Amirpourazarian, Christian Timmerer, Mario Taschwer, and Klaus Schöffmann. 2020. Relevance-Based Compression of Cataract Surgery Videos Using Convolutional Neural Networks. In *Proceedings of the 28th ACM International Conference on Multimedia*. ACM, Seattle WA USA, 3577–3585. doi:10.1145/3394171.3413658
- [2] M. H. Izadimehr, Milad Ghanbari, Guodong Chen, Wei Zhou, Xiaoshuai Hao, Mallesham Dasari, Christian Timmerer, and Hadi Amirpour. 2025. SVD: Spatial Video Dataset. doi:10.48550/ARXIV.2506.06037 Version Number: 1.
- [3] Marc Lambooi, Wijnand Ijsselstein, Don G. Bouwhuis, and Ingrid Heynderickx. 2011. Evaluation of Stereoscopic Images: Beyond 2D Quality. *IEEE Transactions on Broadcasting* 57, 2 (June 2011), 432–444. doi:10.1109/TBC.2011.2134590
- [4] Kyoung Won Nam, Jeongyun Park, In Young Kim, and Kwang Gi Kim. 2012. Application of Stereo-Imaging Technology to Medical Field. *Healthcare Informatics Research* 18, 3 (2012), 158. doi:10.4258/hir.2012.18.3.158
- [5] David O’Brart. 2025. The future of cataract surgery. *Eye* (March 2025). doi:10.1038/s41433-025-03745-x
- [6] Manfred Jürgen Primus, Doris Putzgruber-Adamitsch, Mario Taschwer, Bernd Münzer, Yosuf El-Shabrawi, Laszlo Böszörményi, and Klaus Schoeffmann. 2018. Frame-Based Classification of Operation Phases in Cataract Surgery Videos. In *MultiMedia Modeling*, Klaus Schoeffmann, Thanarat H. Chalidabhongse, Chong Wah Ngo, Supavadee Aramvith, Noel E. O’Connor, Yo-Sung Ho, Moncef Gabbouj, and Ahmed Elgammal (Eds.). Vol. 10704. Springer International Publishing, Cham, 241–253. doi:10.1007/978-3-319-73603-7\_20 Series Title: Lecture Notes in Computer Science.
- [7] Katie Schwab, Ralph Smith, Vanessa Brown, Martin Whyte, and Iain Jourdan. 2017. Evolution of stereoscopic imaging in surgery and recent advances. *World Journal of Gastrointestinal Endoscopy* 9, 8 (2017), 368. doi:10.4253/wjge.v9.i8.368
- [8] Gerhard Tech, Ying Chen, Karsten Muller, Jens-Rainer Ohm, Anthony Vetro, and Ye-Kui Wang. 2016. Overview of the Multiview and 3D Extensions of High Efficiency Video Coding. *IEEE Transactions on Circuits and Systems for Video Technology* 26, 1 (Jan. 2016), 35–49. doi:10.1109/TCSVT.2015.2477935
- [9] Jeroen Van Der Hooft, Hadi Amirpour, Maria Torres Vega, Yago Sanchez, Raimund Schatz, Thomas Schierl, and Christian Timmerer. 2023. A Tutorial on Immersive Video Delivery: From Omnidirectional Video to Holography. *IEEE Communications Surveys & Tutorials* 25, 2 (2023), 1336–1375. doi:10.1109/COMST.2023.3263252
- [10] Ethan Waisberg, Joshua Ong, Mouayad Masalkhi, Nasif Zaman, Prithul Sarker, Andrew G. Lee, and Alireza Tavakkoli. 2024. Apple Vision Pro and why extended reality will revolutionize the future of medicine. *Irish Journal of Medical Science (1971 -)* 193, 1 (Feb. 2024), 531–532. doi:10.1007/s11845-023-03437-z
- [11] Ethan Waisberg, Joshua Ong, Mouayad Masalkhi, Nasif Zaman, Prithul Sarker, Andrew G. Lee, and Alireza Tavakkoli. 2024. The future of ophthalmology and vision science with the Apple Vision Pro. *Eye* 38, 2 (Feb. 2024), 242–243. doi:10.1038/s41433-023-02688-5
- [12] Wei Zhou, Ning Liao, Zhibo Chen, and Weiping Li. 2016. 3D-HEVC visual quality assessment: Database and bitstream model. In *2016 Eighth International Conference on Quality of Multimedia Experience (QoMEX)*. IEEE, Lisbon, Portugal, 1–6. doi:10.1109/QoMEX.2016.7498946
- [13] Wei Zhou, Hadi Amirpour, Christian Timmerer, Guangtao Zhai, Patrick Le Callet, and Alan C. Bovik. 2025. Perceptual Visual Quality Assessment: Principles, Methods, and Future Directions. doi:10.48550/ARXIV.2503.00625 Version Number: 1.
- [14] Wei Zhou and Zhou Wang. 2024. Perceptual Depth Quality Assessment of Stereoscopic Omnidirectional Images. *IEEE Transactions on Circuits and Systems for Video Technology* 34, 12 (Dec. 2024), 13452–13462. doi:10.1109/TCSVT.2024.3449696